

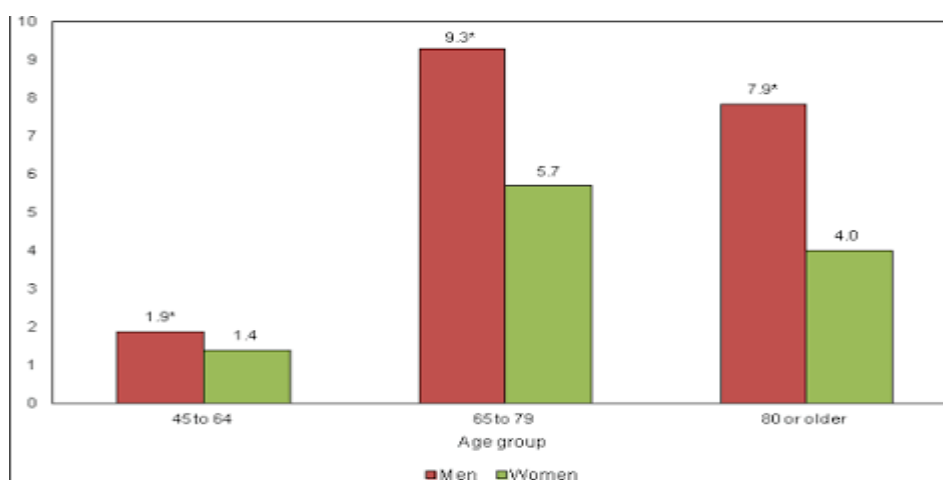
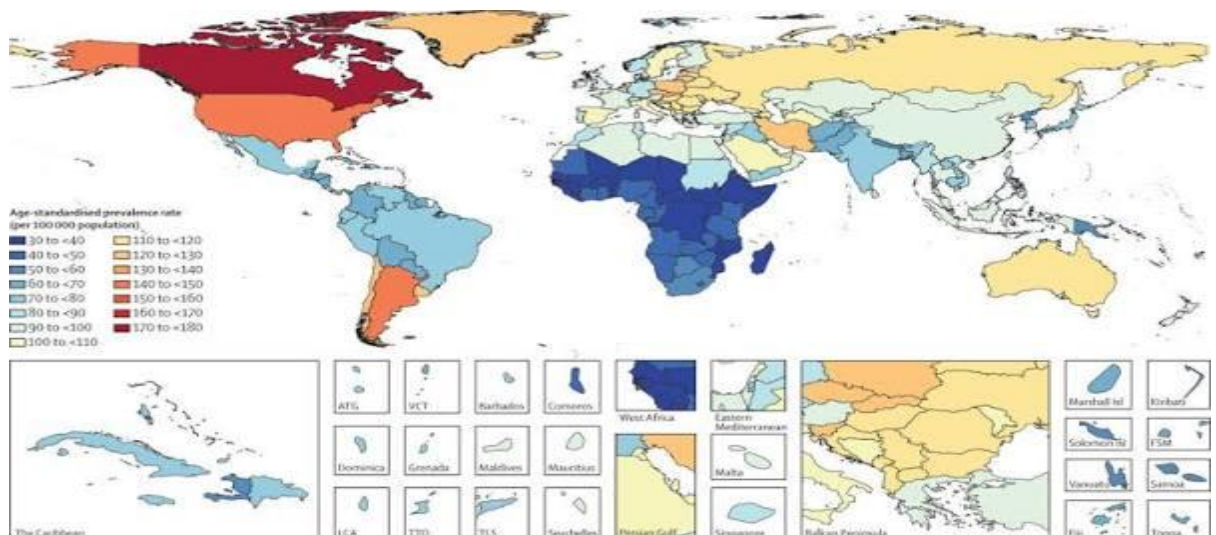
# PARKINSON'S DISEASE

## 1. Definition

Parkinson's disease is a chronic, progressive neurodegenerative disorder primarily affecting the motor system. It results from the selective loss of dopamine-producing neurons in the substantia nigra and the accumulation of  **$\alpha$ -synuclein** protein aggregates (Lewy bodies).

## 2. Epidemiology

- Second most common neurodegenerative disorder
- Affects approximately 1% of people over 60 years
- More common in men than women
- Average age of onset: 60 years
- Early-onset Parkinson's disease can occur before age 50



Source: Statistics Canada, CANSIM Table 105-1305.

# 3. Risk Factors

## Non-modifiable

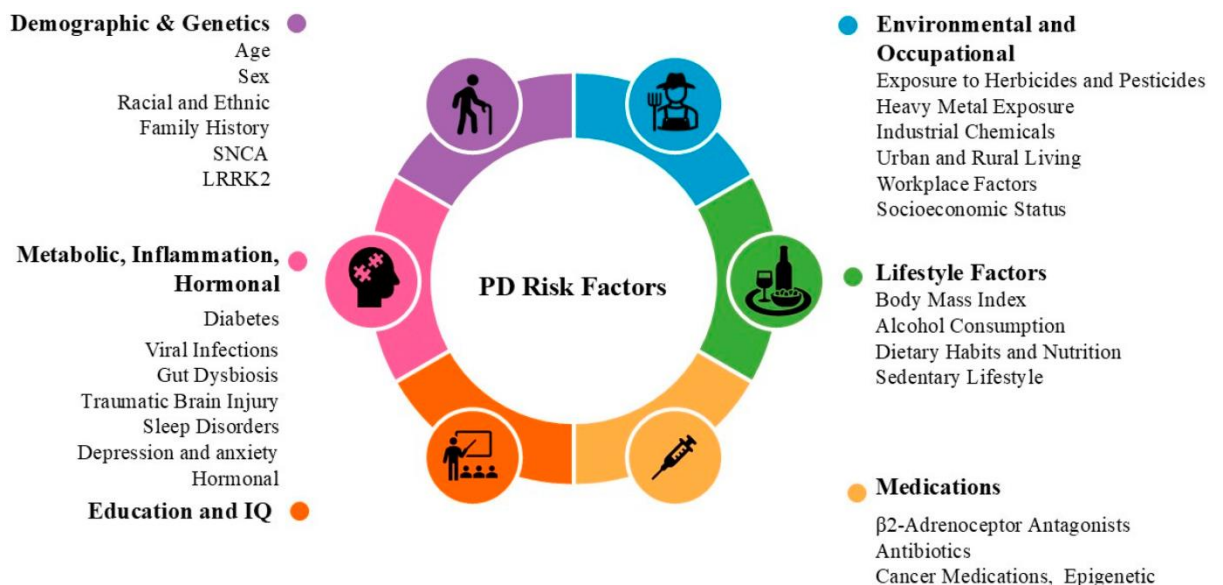
- Increasing age
- Family history
- Male sex
- Genetic mutations

## Modifiable

- Pesticide exposure
- Herbicides
- Heavy metals
- Head injury
- Rural living
- Air pollution

Protective factors (associated in some studies)

- Regular exercise
- Coffee/caffeine consumption



# 4. Brain Regions Affected

## Substantia Nigra Pars Compacta

Primary site of dopaminergic neuron degeneration.

## **Basal Ganglia**

Includes:

- Caudate nucleus
- Putamen
- Globus pallidus

Responsible for motor control.

## **Striatum**

Receives dopamine from substantia nigra.

Loss of dopamine causes impaired movement.

## **Thalamus**

Relays motor information to the cerebral cortex.

## **Motor Cortex**

Controls voluntary movement.

## **Brainstem**

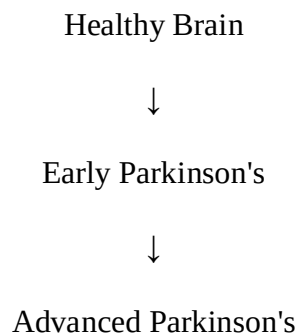
Contains nuclei involved in sleep and autonomic regulation.

## **Later Disease Stages**

Degeneration extends to:

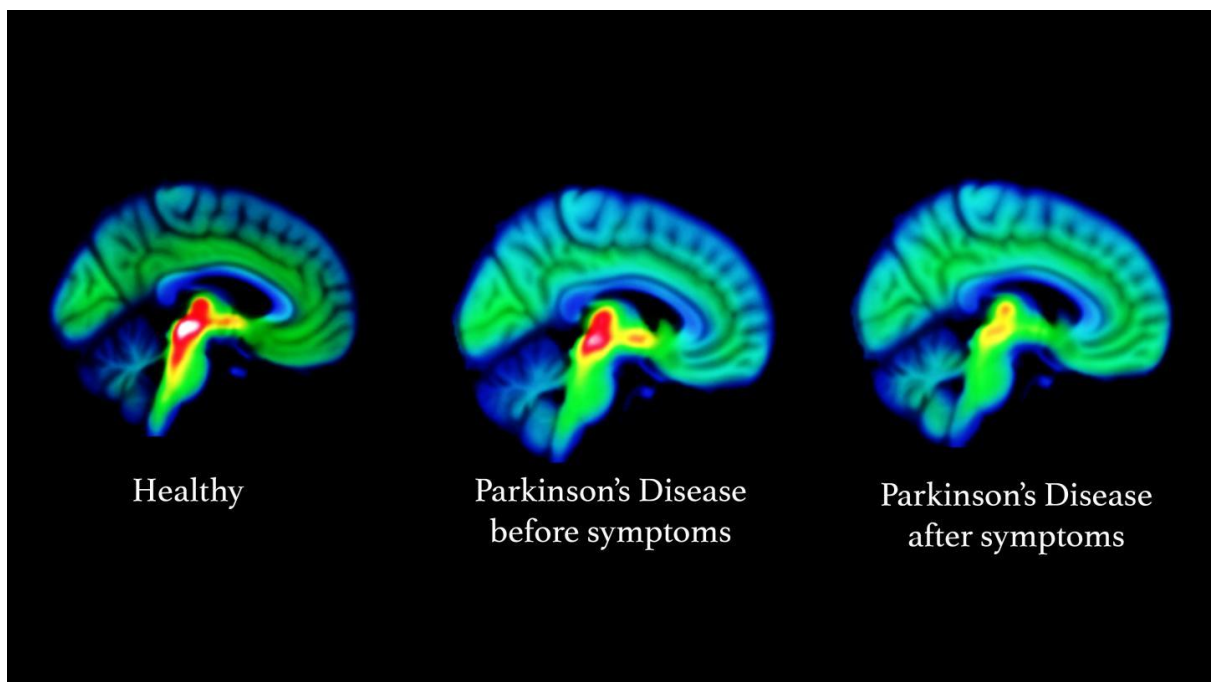
- Cortex
- Limbic system
- Autonomic nervous system

## 5. Anatomy: Healthy vs Parkinson's Brain



Key changes:

- Loss of pigmented neurons in substantia nigra
- Reduced dopamine levels
- Presence of Lewy bodies
- Changes in basal ganglia circuitry



## 6. Pathophysiology

### Dopamine Loss

Degeneration of dopaminergic neurons



Reduced dopamine release



Impaired basal ganglia signaling



Motor dysfunction

## **$\alpha$ -Synuclein Aggregation**

Normal  $\alpha$ -synuclein



Misfolding



Aggregation



Lewy bodies



Neuronal dysfunction



Cell death

## **Mitochondrial Dysfunction**

Reduced ATP production



Oxidative stress



Neuronal damage

## Neuroinflammation

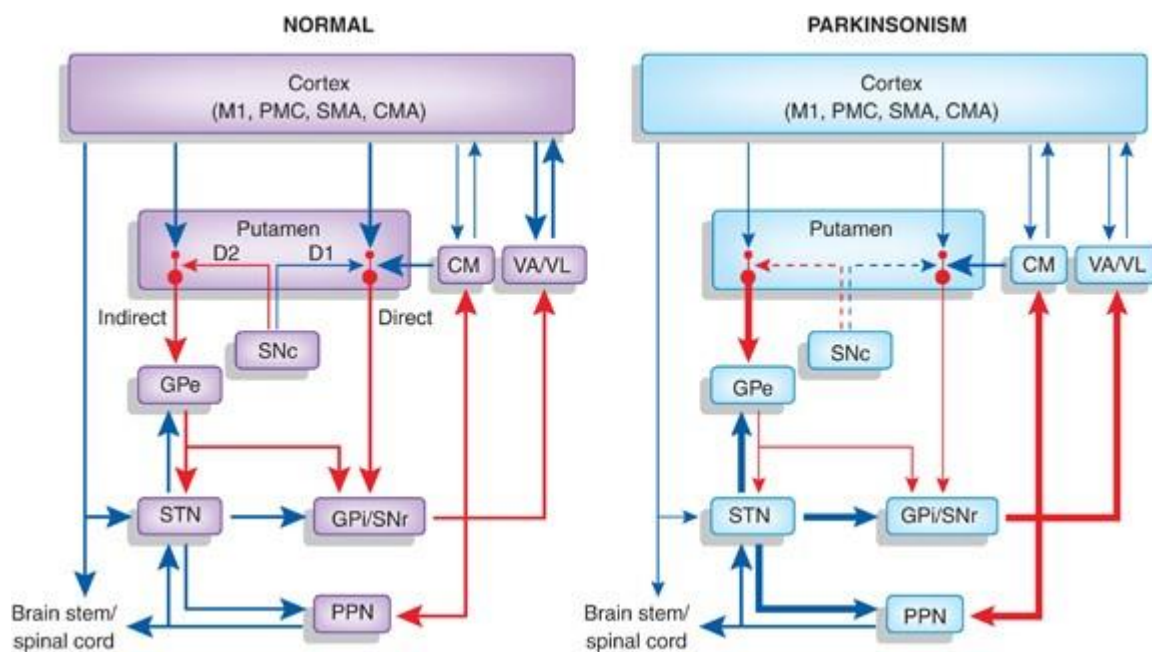
Activation of:

- Microglia
- Astrocytes

Release of inflammatory cytokines contributing to disease progression.

## Oxidative Stress

- Reactive oxygen species (ROS)
- Lipid peroxidation
- DNA damage



## 7. Genetics

Important genes:

### SNCA

Encodes  $\alpha$ -synuclein.

Mutations cause abnormal protein aggregation.

## **LRRK2**

Most common genetic cause of familial Parkinson's disease.

## **PARK2 (Parkin)**

Associated with early-onset Parkinson's disease.

## **PINK1**

Maintains mitochondrial quality control.

## **DJ-1**

Protects against oxidative stress.

## **GBA**

Mutations increase Parkinson's disease risk.

# **8. Symptoms**

## **Motor Symptoms**

- Resting tremor
- Bradykinesia
- Rigidity
- Postural instability
- Shuffling gait
- Freezing of gait
- Reduced facial expression (masked facies)
- Micrographia (small handwriting)

## **Non-Motor Symptoms**

- Depression
- Anxiety
- Constipation
- Sleep disorders
- Loss of smell (hyposmia)
- Fatigue
- Cognitive impairment
- Dementia (later stages)

## **Autonomic Symptoms**

- Orthostatic hypotension
- Urinary dysfunction
- Excessive sweating
- Sexual dysfunction

# **9. Clinical Stages**

## **Prodromal Stage**

- Loss of smell
- REM sleep behavior disorder
- Constipation

## **Early Parkinson's Disease**

- Mild unilateral symptoms
- Minimal disability

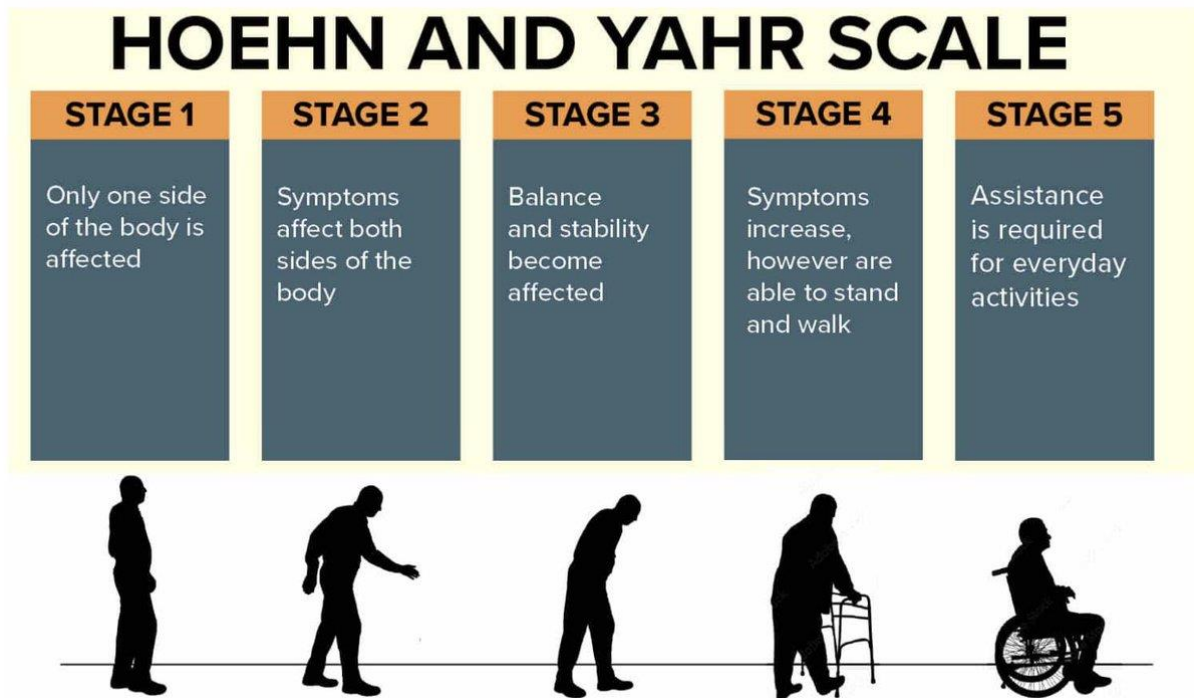
## **Moderate Disease**

- Bilateral symptoms
- Balance impairment



## Advanced Disease

- Severe motor disability
- Cognitive decline
- Dependence on caregivers



## 10. Diagnosis

### Clinical Examination

Neurological assessment

Medical history

Motor examination

### Diagnostic Criteria

- Bradykinesia (required)
- Resting tremor
- Rigidity

## Rating Scales

- Unified Parkinson's Disease Rating Scale (UPDRS)
- Hoehn and Yahr Scale
- MDS-UPDRS

# 11. Neuroimaging

## MRI

Usually normal in early Parkinson's disease.

Used to exclude other causes.

## DaTscan (Dopamine Transporter Scan)

Shows reduced dopamine transporter activity.

## PET

Evaluates dopamine metabolism.

## Transcranial Ultrasound

May show increased echogenicity of substantia nigra.

# 12. Biomarkers

Current research biomarkers include:

- $\alpha$ -Synuclein in CSF
- Neurofilament Light Chain (NfL)
- DJ-1 protein
- Blood inflammatory markers

No single biomarker is currently definitive for routine diagnosis.

## **13. Treatment**

### **Medications**

#### **Dopamine Replacement**

- Levodopa + Carbidopa

#### **Dopamine Agonists**

- Pramipexole
- Ropinirole
- Rotigotine

#### **MAO-B Inhibitors**

- Selegiline
- Rasagiline
- Safinamide

#### **COMT Inhibitors**

- Entacapone
- Opicapone

#### **Amantadine**

Used for dyskinesia.

### **Surgical Treatment**

#### **Deep Brain Stimulation (DBS)**

Targets:

- Subthalamic nucleus (STN)
- Globus pallidus internus (GPi)

### **Non-pharmacological Therapy**

- Physiotherapy
- Occupational therapy
- Speech therapy
- Exercise programs
- Nutritional support

## 14. Latest Research

- $\alpha$ -Synuclein-targeted immunotherapy
- Stem cell transplantation
- Gene therapy
- AI-assisted diagnosis using gait analysis
- Wearable devices for disease monitoring
- Biomarker discovery using blood and CSF
- Neuroprotective drug trials

## 15. Future Directions

- Disease-modifying therapies
- Precision medicine
- Gene editing (CRISPR)
- Stem cell replacement therapy
- AI-based early diagnosis
- Digital biomarkers
- Personalized rehabilitation
- Brain-computer interfaces

## 16. References

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- Nature Reviews Neurology
- Nature Neuroscience
- Neuron
- The Lancet Neurology Movement Disorders
- Parkinsonism & Related Disorders
- National Institute of Neurological Disorders and Stroke (NINDS)
- World Health Organization (WHO)